Daniel Köstler

B01:

a) $U_{Eff} = \frac{\hat{U}}{\sqrt{2}}$

$\hat{U} = \sqrt{2}\, U_{Eff}$

$\hat{U}_{1=2=3} = \sqrt{2} \cdot U_{1=2=3\,Eff} = \sqrt{2} \cdot 230$

$\hat{U}_{1=2=3} = 325,27V$

b) $I_{1,Eff} = \frac{U_{1,eff}}{|Z|}$ $\quad |Z| = R$

$I_{1,Eff} = \frac{230}{23} = 10A = I_{2,Eff} = I_{3,Eff}$

$\hat{I}_1 = \sqrt{2} \cdot I_{1,eff} = \sqrt{2} \cdot 10$

$\hat{I}_1 = 14,14A = \hat{I}_2 = \hat{I}_3$

c) $P_{1=2=3} = U_{Eff} \cdot I_{Eff}$

$P = 2300W$

$P_{ges} = 3 \cdot 2300 = 6900W$

$$I_g = \hat{I} \cdot \left(e^{j\omega t} + e^{j(\omega t + \frac{2\pi}{3})} + e^{j(\omega t + \frac{4\pi}{3})}\right)$$

$$I_g = \hat{I}\, e^{j\omega t} \cdot \left(1 + e^{j\frac{2\pi}{3}} + e^{j\frac{4\pi}{3}}\right)$$

$$I_g = 0$$

Da die einzelnen Ströme um jeweils 120° phasenverschoben sind, es sich um ein Drehstromsystem handelt und es an jedem Leiter eine gleich große ohmsche Last hat, fließt durch den Nullleiter kein Strom.
Daher ist die Stellung des Schalters egal.

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B02:

a)

$a_1)\ Z_1 = R_1 + j\omega L_1$

$$|Z_1| = \sqrt{R_1^2 + (\omega L_1)^2} = \sqrt{29^2 + (2\pi 50 \cdot 0{,}099)^2} = 42{,}52\,\Omega$$

$$Z_2 = R_2 + j\cdot\left(\omega L_2 - \frac{1}{\omega C_2}\right)$$

$$|Z_2| = \sqrt{R_2^2 + \left(\omega L_2 - \frac{1}{\omega C_2}\right)^2} = \sqrt{40^2 + \left(2\pi 50 \cdot 0{,}127 - \frac{1}{2\pi\, 50 \cdot 24 \cdot 10^{-6}}\right)^2}$$

$$|Z_2| = 100{,}99\,\Omega$$

$$|Z_{12}| = \frac{|Z_1| \cdot |Z_2|}{|Z_1| + |Z_2|} = \frac{42{,}52 \cdot 100{,}99}{42{,}52 + 100{,}99} = 29{,}92\,\Omega$$

$a_2)\ \varphi_1 = \arctan\left(\frac{\omega L_1}{R_1}\right) = 47^\circ$

$\varphi_2 = \arctan\left(\frac{\omega L_2 - \frac{1}{\omega C_2}}{R_2}\right) = -66{,}67^\circ$

Winkel von Phase

Phasenverschiebung Strom 1: $\varphi_1 = -47^\circ$

—//— Strom 2: $\varphi_2 = 66{,}67^\circ$

Phasenunterschied gesamt: $113{,}67^\circ$

$a_3)$ $I_{eff} = \frac{U_{eff}}{|Z_{12}|} = \frac{230}{29{,}92} = 7{,}69\,A$

$P_S = U_{eff} \cdot I_{eff} = 1768{,}7\,VA$

$P_W = P_S \cdot \cos(113{,}67) = -709{,}79\,W$

$P_B = P_S \cdot \sin(113{,}67) = 1619{,}9\,VAR$

$a_4)$ $\frac{U_{L2,eff}}{U_{eff}} = \frac{|Z_{L2}|}{|Z_2|}$

$U_{L2,eff} = U_{eff} \cdot \frac{|Z_{L2}|}{|Z_2|}$

$Z_{L2} = j\omega L_2$

$|Z_{L2}| = \omega L_2 = 39{,}89\,\Omega$

$U_{L2,eff} = 230 \cdot \frac{39{,}89}{100{,}99} = 90{,}85\,V$

$\hat{U}_{L2} = \sqrt{2} \cdot U_{L2,eff} = 128{,}48\,V$

b) Die Hilfswicklung mit dem Kondensator muss gedreht werden.

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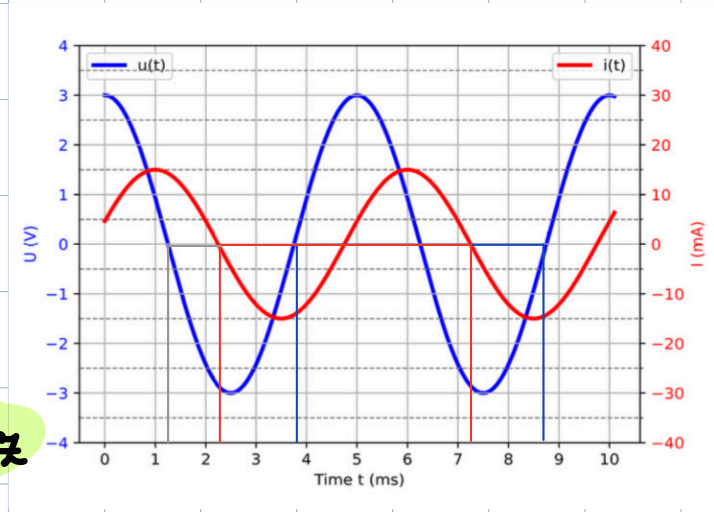


B03:

a) $f_i = \frac{1}{T_i} = \frac{1}{5 \cdot 10^{-3}} = 200\,Hz$

$f_U = \frac{1}{T_U} = \frac{1}{5 \cdot 10^{-3}} = 200\,Hz$

$\omega = 2\pi f = 400\pi$

$\Delta t = 1ms$

$T = 5ms$

$\frac{\varphi}{360°} = \frac{\Delta t}{T}$

$\varphi = 360 \cdot \frac{\Delta t}{T} = 72°$ $(\varphi_i = -72°)$ (weil Strom eilt nach: aber nur Betrag wichtig)

b) $\hat{U} = 3V$, $\hat{I} = 15mA = 1{,}5 \cdot 10^{-2} A$

$U_{eff} = \frac{\hat{U}}{\sqrt{2}} = \frac{3}{\sqrt{2}} V$

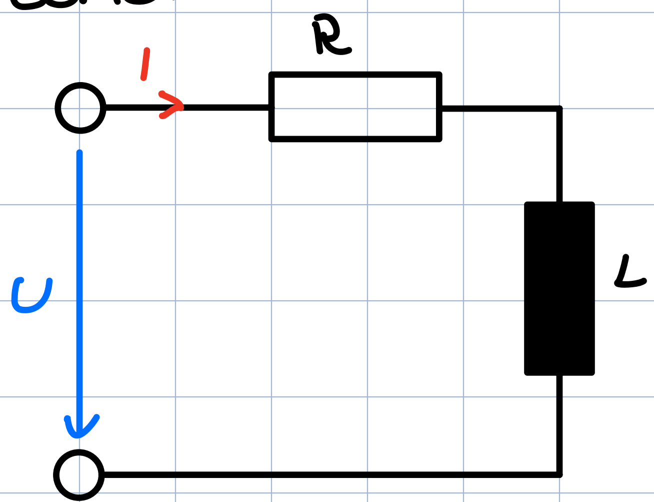
$I_{eff} = \frac{\hat{I}}{\sqrt{2}} = \frac{1{,}5 \cdot 10^{-3}}{\sqrt{2}} A$

$\frac{U_{eff}}{I_{eff}} = |Z| = 2000\Omega$

$P_B = I_{eff} \cdot U_{eff} \cdot \sin(72) = 0{,}021W = 21mW$

c) Strom eilt Spannung nach $\Rightarrow$ Induktivität

Serie:



I:

$$\underline{Z} = R + j\omega L$$
$$|\underline{Z}| = \sqrt{R^2 + \omega^2 L^2}$$
$$U_{eff} = |\underline{Z}| \cdot I_{eff}$$
$$\frac{U_{eff}}{I_{eff}} = \sqrt{R^2 + \omega^2 L^2}$$
$$\frac{U_{eff}^2}{I_{eff}^2} - R^2 = \omega^2 L^2$$
$$L = \sqrt{\frac{\frac{U_{eff}^2}{I_{eff}^2} - R^2}{\omega^2}}$$

II:

$$\arctan\left(\frac{\omega L}{R}\right) = 72^\circ$$
$$\tan(72) = \frac{\omega L}{R}$$
$$L = \frac{\tan(72) \cdot R}{\omega}$$

$$\frac{\sqrt{|Z|^2 - R^2}}{\omega} = \frac{\tan(72) \cdot R}{\omega}$$

$$|Z|^2 - R^2 = \tan^2(72) \cdot R^2$$
$$|Z|^2 = R^2 \cdot (1 + \tan^2(72))$$
$$R = \frac{|Z|}{\sqrt{1 + \tan^2(72)}} = \frac{2000}{\sqrt{1 + \tan^2(72)}} = 618\,\Omega$$

$$L = \sqrt{\frac{\frac{U_{eff}^2}{I_{eff}^2} - R^2}{\omega^2}} = 1{,}51\,H$$

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B04:

a) $\frac{U_1}{U_2} = \frac{N_1}{N_2}$

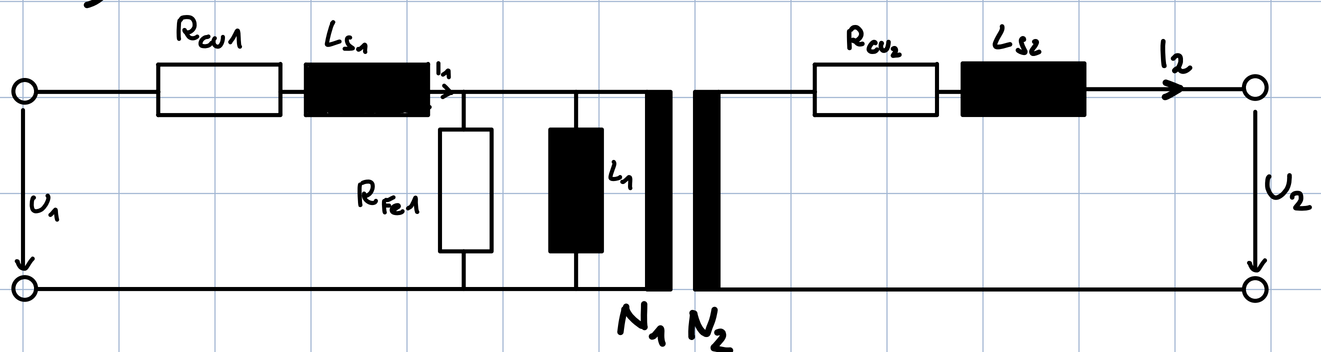
$N_2 \cdot \frac{U_1}{U_2} = N_1$

$N_2 = N_1 \cdot \frac{U_2}{U_1} = 200 \cdot \frac{24}{230} = 20{,}87 \approx 21$

b) $\frac{N_1}{N_2} = \frac{U_1}{U_2} = \frac{I_2}{I_1}$

$I_2 = \frac{U_1}{U_2} \cdot I_1 = \frac{230}{24} \cdot 1 = 9{,}58\,A$

c)



c_1) Leerlauf: kein Strom ($i_2 = 0$)

Relevant:

Hauptinduktivitäten

Eisenverluste

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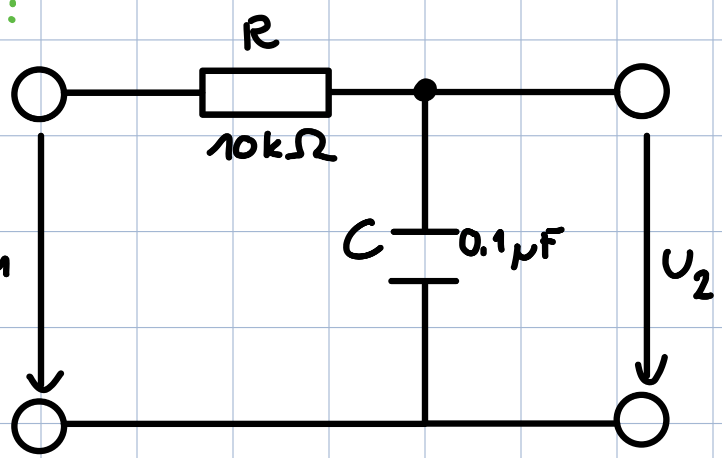
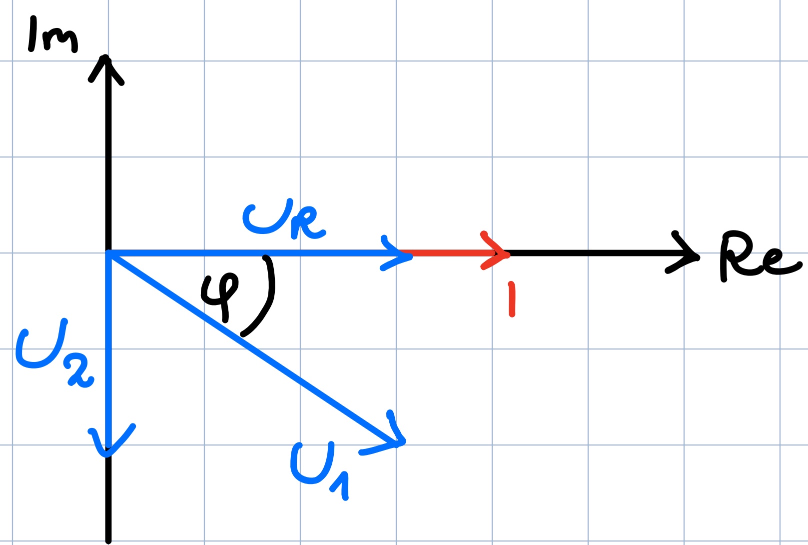
B05:

a)



b)



$$U_R = I \cdot R$$

$$U_C = I \cdot \frac{1}{j\omega C} = -I \cdot \frac{j}{\omega C}$$

c) $T = 20\,ms$

$$f = \frac{1}{20 \cdot 10^{-3}} = 50\,Hz$$

$$g(\omega) = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R} = \frac{1}{1 + j\omega RC}$$

$$|g(\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + (2\pi \cdot 50 \cdot 10^4 \cdot 0{,}1 \cdot 10^{-6})^2}}$$

$|g(\omega)| = 0{,}954$

$\varphi(\omega) = -\arctan(\omega RC) = -17{,}44°$

$g[dB] = 20 \cdot \log(|g(\omega)|) = -0{,}41 dB$

Serienschaltung: Strom überall gleich groß

$|Z| = \sqrt{R^2 + (\frac{1}{\omega C})^2} = 33{,}364 k\Omega$

$U_1 = |Z| \cdot I$

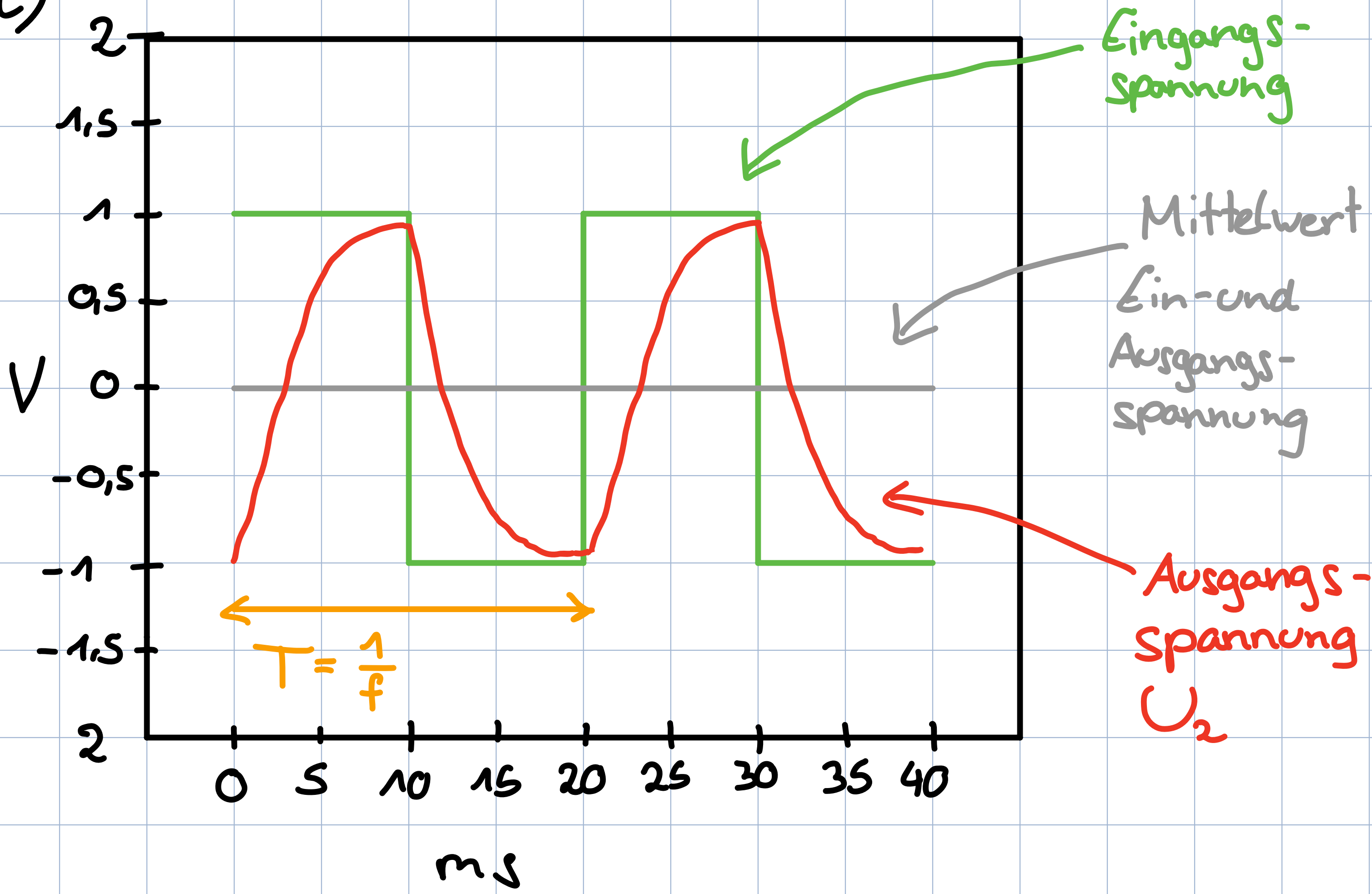
$I = \frac{U_1}{|Z|} = \frac{10}{33{,}364 \cdot 10^3} \doteq 0{,}0003 A$

$U_2 = \sqrt{U_1^2 - U_R^2}$ (Satz des Pythagoras)

$U_2 = \sqrt{U_1^2 - (I \cdot R)^2}$

$U_2 = \sqrt{10^2 - (0{,}0003 \cdot 10^4)^2} = 9{,}54 V$

d)



2
1,5
1
0,5
V
0
-0,5
-1
-1,5
2
0
5
10
15
20
25
30
35
40
ms
$T = \frac{1}{f}$
Eingangs-spannung
Mittelwert Ein- und Ausgangs-spannung
Ausgangs-spannung
U_2

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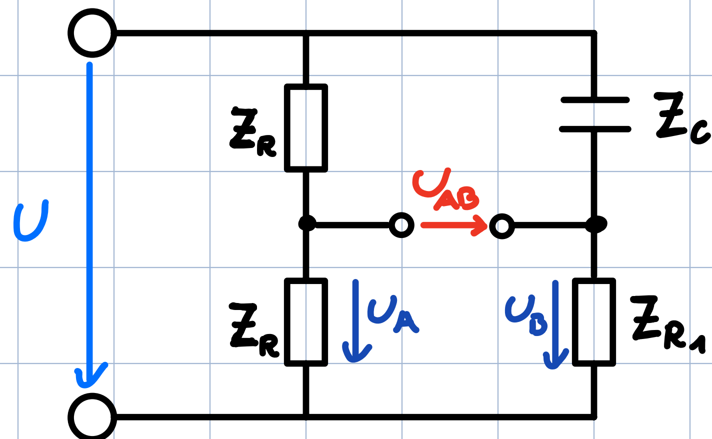

B06:

a) $Z_C = \frac{1}{j\omega C}$

$|Z_C| = \frac{1}{\omega C} = \frac{1}{2\pi 50 \cdot 10^{-6}} = 3183\,\Omega$

$$\frac{Z_R}{Z_C} = \frac{Z_R}{Z_{R1}}$$

$Z_{R1} = Z_C$

$Z_{R1} = 3183\,\Omega$

b) $\frac{U_{eff}}{U_{AB,eff}} =$

c) $$\frac{1}{Z} = \frac{1}{2R} + \frac{1}{R_1 - \frac{j}{\omega C}}$$

$$Y = \frac{1}{2R} + \frac{1}{R_1 - \frac{j}{\omega C}}$$

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B08:

a) $g(\omega) = \frac{U_2}{U_1}$

$$Z_{ges} = R + j\omega L + \frac{1}{j\omega C} = R + j\cdot\left(\omega L - \frac{1}{\omega C}\right)$$
$$Z_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C}$$

$$\omega_0 L = \frac{1}{\omega_0 C}$$
$$\omega_0 = \frac{1}{\sqrt{LC}} = 10^5 Hz$$ Resonanz(kreis)frequenz

$$f_g = \frac{1}{2\pi\sqrt{LC}}$$ Grenzfrequenz

$$X_C = \frac{1}{\omega C},\quad X_L = \omega L$$

$$g(\omega) = \frac{U_2}{U_1} = \frac{-jX_C}{-jX_C + jX_L + R} = \frac{X_C}{X_C - X_L + jR}$$

$$g(\omega) = \frac{X_C\cdot(X_C - X_L - jR)}{(X_C - X_L + jR)\cdot(X_C - X_L - jR)} = \frac{X_C^2 - X_C X_L - jRX_C}{(X_C - X_L)^2 + R^2}$$

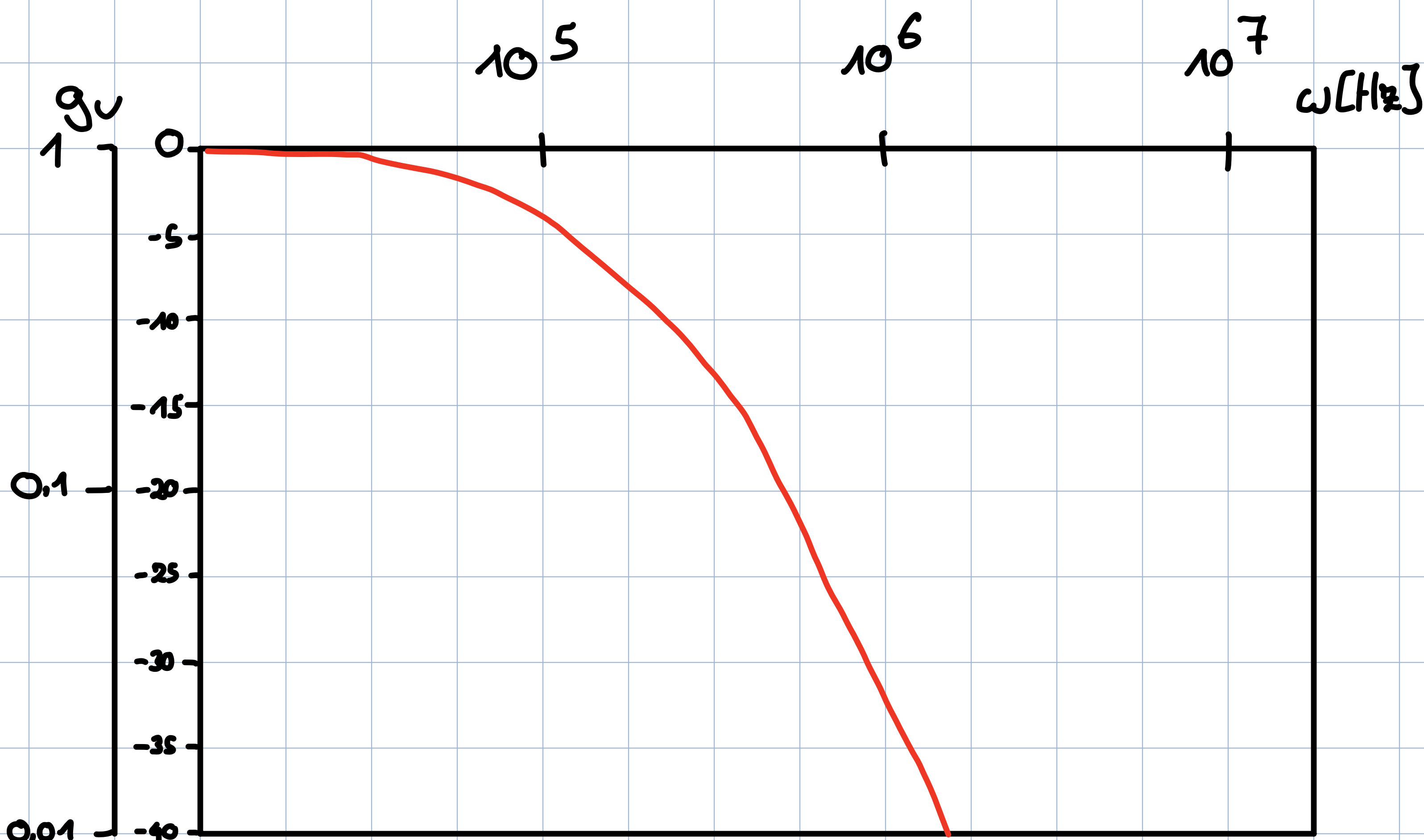
$$g(\omega) = \frac{X_C^2 - X_C X_L}{(X_C - X_L)^2 + R^2} - j\,\frac{R \cdot X_C}{(X_C - X_L)^2 + R^2}$$

$$|g(\omega)| = \sqrt{\frac{\left(\left(\frac{1}{\omega C}\right)^2 - \frac{L}{C}\right)^2 + \left(\frac{R}{\omega C}\right)^2}{\left(\frac{1}{\omega C} - \omega L\right)^2 + R^2}}$$

$$|g(\omega_0)| = 10$$

b) $\omega_g = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10^{-10}}} \approx 1{,}6 \cdot 10^4$

$$x = 5 \cdot \log\left(\frac{\frac{1}{2\pi\sqrt{10^{-10}}}}{10^4}\right) \approx 1\,\text{cm}$$



c) Dabei handelt es sich um einen Tiefpass. Das liegt daran, dass der Filter nur niedrige Frequenzen durchlässt und die Ausgangsspannung am Kondensator abgegriffen wird.